5] A Certain Fluid at 10 bar is Contained in a Cylinder behind a Piston, the intial Value being 0.05 m3. Calculate the work done by the fluid when it expands reversibly:

i. At Constant pressure to a final volume of 0.2 m3

ii. According to a law pV = Constant to a final volume of 0.1 m3

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iii. According to a law $pv^3 = C$ to find volume of $0.06 m^3$

iv. stretch all processes on a p-v diagram.

(i)

$P_1 = 10 \times 10^5$	expand	$P_2 = 10 \times 10^5$
$V_1 = 0.05 m^3$	$P = C$ $P_1 = P_2$	$V_2 = 0.2 m^3$

$$W = P(V_2 - V_1) = 10 \times 10^5 (0.2 - 0.05) = 15 \times 10^4 J$$

P_1 P_2

(ii)

$P_1 = 10 \times 10^5$	expand	P_2
$V_1 = 0.05 m^3$	$PV = C$ $P_1V_1 = P_2V_2 = C$	$V_2 = 0.1 m^3$

$$W = P_1 V_1 \ln \frac{V_2}{V_1} = 10 \times 10^5 \times 0.05 \times \ln \frac{0.1}{0.05} = 34657.36 J$$

(iii)

$P_1 = 10 \times 10^5$	expand	P_2
$V_1 = 0.05 m^3$	$PV^3 = C$ $P_1V_1^3 = P_2V_2^3$	$V_2 = 0.06 m^3$

$$W = \frac{P_1V_1 - P_2V_2}{n-1} =$$

$P_1 V_1^3 = P_2 V_2^3$ $P_2 =$ Pascal

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6] $0.05\ m^3$ of a gas at 6.9 bar expands reversibly in a cylinder behind a piston according to the law $pv^{1.2} = C$, until the volume is $0.08\ m^3$. Calculate the work done by the gas and sketch the process on a p-V diagram.

p_2, $W = ??$

State (1)		State (2)
$V_1 = 0.05\ m^3$	expand $\xrightarrow{pv^{1.2} = C}$	$V_2 = 0.08\ m^3$
$P_1 = 6.9 \times 10^5$	$P_1 V_1^{1.2} = P_2 V_2^{1.2} = C$	$P_2 = 3.9\ bar$

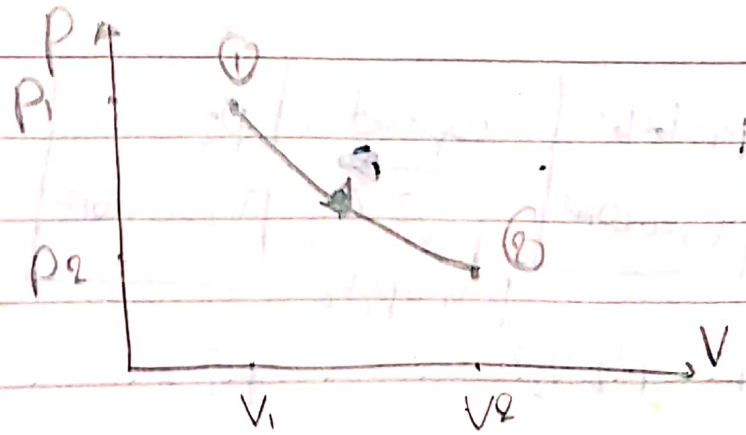
$$P_1 V_1^{1.2} = P_2 V_2^{1.2}$$

$$6.9 \times 10^5 \times (0.05)^{1.2} = P_2 (0.08)^{1.2}$$

$$P_2 = 6.9 \times 10^5 \left(\frac{0.05}{0.08}\right)^{1.2}$$

$$P_2 = 3.9 \times 10^5\ pascal$$

$$W = \frac{P_1 V_1 - P_2 V_2}{n-1} = 16500\ joul$$



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